CLIMATE ANALYSIS REPORT

European Cities Are Running About 1 °C Warmer Than Their 1991–2020 Normal

Temperature anomalies 2021–2024 for 113 cities with 500,000+ people, from ERA5 and Overture Maps

Recent period 2021–2024 · Baseline 1991–2020 (WMO) · Data: Copernicus ERA5 via Planette, Overture Maps

+1.0 °C	172M	78.7M	+1.6 °C
Mean city anomaly, 2021–2024	People in 113 cities, 30 countries	People in 62 cities at or above +1.0 °C	Highest city: Bucharest

1 Summary

How much warmer were Europe's large cities in 2021–2024 than in the 1991–2020 WMO baseline, and how many people live in them? Across **113 cities in 30 countries (172 million people)**, the mean anomaly is **+1.0 °C**. **62 cities with 78.7 million people** are at or above +1.0 °C. Warming is strongest in the east and southeast: Bucharest leads at **+1.6 °C**, and the 2024 land anomaly in eastern Europe is about twice that in western Europe.

2 Key results

- **Range across cities:** from **+0.3 °C** (Oslo) to **+1.6 °C** (Bucharest). Bucharest is the only city at or above +1.5 °C; no city reaches +2.0 °C.
- **2024 alone was exceptional:** averaged over all land in the European domain, the 2024 anomaly is **+1.6 °C**.
- **East–west contrast in 2024:** the eastern box (20–40°E, 42–60°N) averages **+2.1 °C** over land, against **+1.0 °C** in the western box (10°W–10°E, 42–60°N), a difference of **+1.2 °C** (Figure 1).

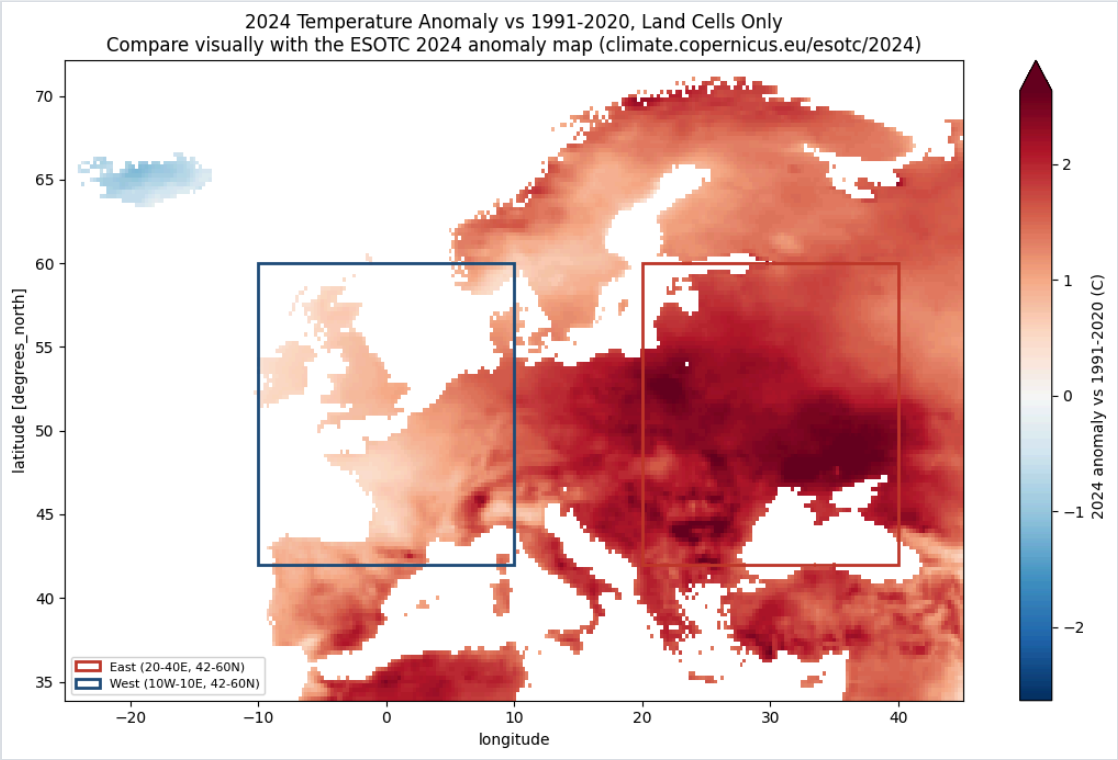


Figure 1. Annual temperature anomaly for 2024 vs 1991–2020 over land, with the east and west comparison boxes. Source: ERA5 (Planette archive); land mask from Overture country polygons.

3 Where it is warming most

- **Highest:** Romania (+1.6 °C), Austria and Serbia (+1.3 °C), Moldova and Ukraine (+1.2 °C).
- **Lowest:** Sweden (+0.5 °C) and Norway (+0.3 °C).
- **Caution:** a country with one city is a single ERA5 grid cell, not a national average; 20 of the 30 countries have fewer than three cities.

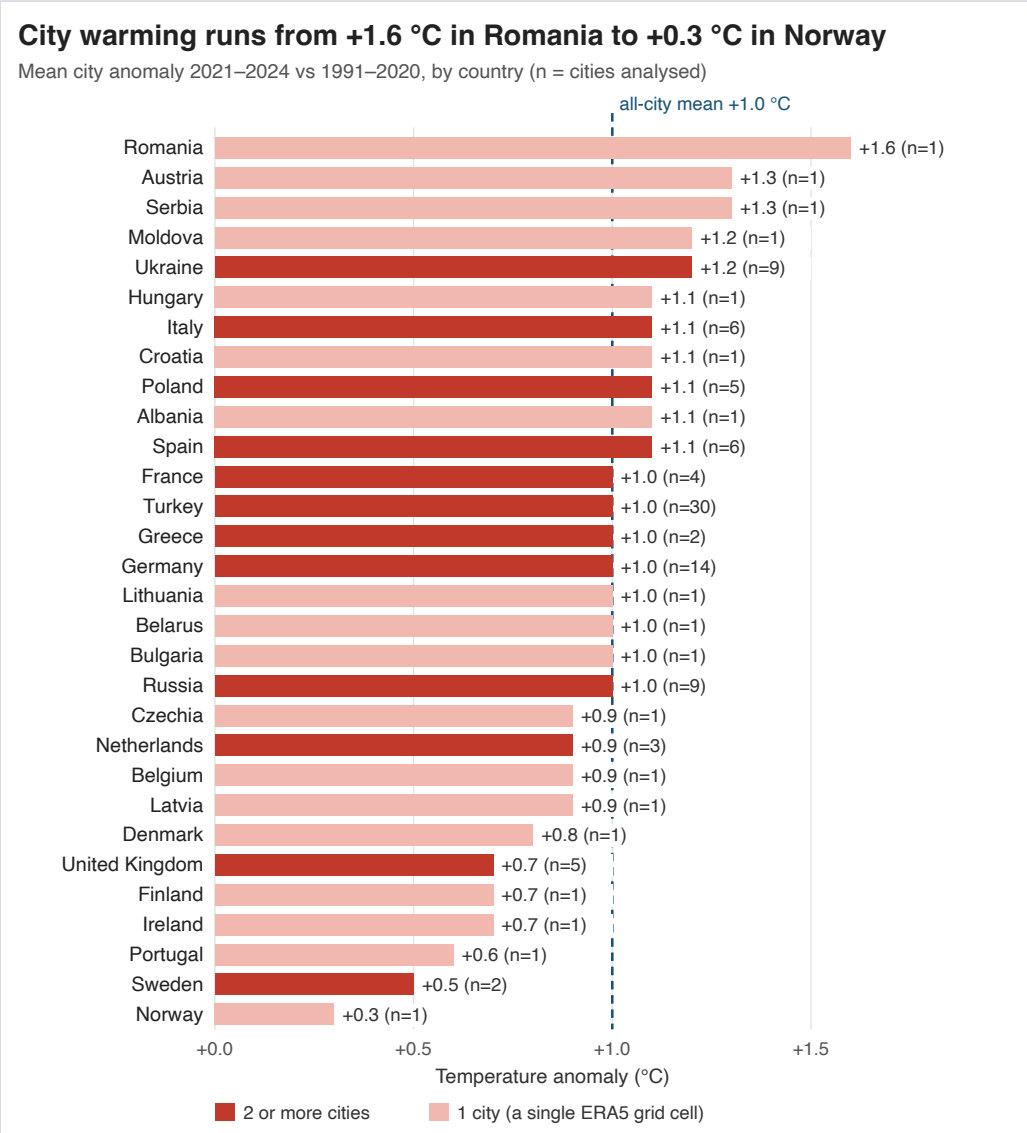


Figure 2. Mean city anomaly 2021–2024 vs 1991–2020 by country; pale bars mark countries with one city. Source: ERA5 (Planette archive), Overture Maps.

4 Population exposure

Cities and people by anomaly band, 2021–2024

ANOMALY BAND	CITIES	PEOPLE
≥ +1.5 °C	1	1.9M
+1.0 to +1.5 °C	61	76.8M
+0.5 to +1.0 °C	50	92.2M
< +0.5 °C	1	1.1M
Total	**113**	**172M**

Most exposed cities (population × anomaly)

CITY	ANOMALY	POPULATION
Istanbul (TR)	+0.9 °C	15.7M
Moscow (RU)	+0.7 °C	12.7M
London (GB)	+0.7 °C	8.9M
Saint Petersburg (RU)	+1.0 °C	5.7M
İzmir (TR)	+1.0 °C	4.5M
Berlin (DE)	+1.1 °C	3.8M

Exposure is driven by population size: the three most exposed cities all sit below the +1.0 °C average.

5 Validation

CHECK	THIS ANALYSIS	PUBLISHED	RESULT
Europe, 2024 land anomaly (ESOTC domain)	+1.6 °C	+1.47 °C (Copernicus ESOTC 2024)	PASS ±0.3 °C
Turkey, 2024 land anomaly	+1.7 °C	+1.70 °C (Turkish State Met. Service)	PASS ±0.3 °C
East vs west, 2024 land means	+2.1 / +1.0 °C	East warmest (ESOTC 2024)	CONSISTENT

- **Sanity check against official WMO 1991–2020 station normals (five cities):** city baselines sit **0.3–1.0 °C below** the mean-temperature normals but **4.2–5.9 °C below** the daily-maximum normals.
- **What this shows:** together with Vienna's July 2024 values (mean 24.2 °C, highest day 28.5 °C), the archive's `t2m_max` variable behaves like daily **mean** temperature, so the analysis labels it `t2m`.

6 Methodology

1. **Cities.** 330,261 Overture `divisions` localities with population in the Europe box are queried with DuckDB directly from S3 GeoParquet; 146 have 500,000+ people.
2. **Cleaning.** Non-European countries and administrative units that are not cities are dropped, as are localities within 1 km of, or inside the boundary of, a more populous one. In Turkey, a metropolitan municipality covers its whole province, so only the most populous locality per province is kept (81 provinces, 20 districts removed). Result: 113 cities.
3. **Temperature.** ERA5 `t2m_max` comes from Planette's monthly store (408 months, 1991–2024), read lazily with Icechunk and xarray over the C3S European domain (25°W–45°E, 34–72°N).
4. **Anomaly.** Anomalies follow the WMO and Copernicus convention: monthly means are compared with a 1991–2020 monthly climatology, and the annual anomaly is the equal-weighted mean of the 12 monthly anomalies (WMO, 2017; C3S). The recent period is 2021–2024.
5. **Extraction and exposure.** One vectorised nearest-grid-cell selection reads every city's values; exposure is the sum of city populations per anomaly band.

7 Performance

STEP (THIS RUN)	TIME
Overture city query	77.3 s
Overture polygons (from cache)	7.1 s
ERA5 connect and open (metadata)	4.5 s
ERA5 anomaly grids (from cache)	1.0 s
Choropleth and land mask	7.0 s
Total, timed steps	**97.0 s**

This run loaded the anomaly grids and polygons from cache. The monthly store holds 12 months per chunk against 31 days for the daily store, so the grid computation reads far less data. A first-run computation time is not recorded in the notebook outputs.

8 Caveats

- `t2m_max` in this archive behaves like daily mean temperature, not daily maximum.
- ERA5 cells are ~28 km: values are regional, exclude the urban heat island, and coastal cells can be partly sea.
- Four years that include the record-warm 2024 describe recent conditions, not a long-term trend.
- Countries with one city are represented by a single grid cell.
- Populations follow administrative definitions; Turkish metropolitan figures cover whole provinces.
- Exposure is a simple population count, with no vulnerability or heat-stress weighting.

9 Data and sources

- Overture Maps Foundation, `divisions` theme, release 2026-09-23.0 (ODbL).
- Planette ERA5 Archive, github.com/PlanetteAI/planette_era5_archive, based on Copernicus ERA5 (CC BY 4.0).
- Copernicus/WMO, *European State of the Climate 2024*: climate.copernicus.eu/esotc/2024
- Turkish State Meteorological Service, *State of the Climate in Turkey in 2024*: mgm.gov.tr/eng/Yearly-Climate/State_of_the_Climate_in_Turkey_in_2024.pdf
- WMO Climate Normals for 1991–2020, NOAA NCEI accession 0253808: ncei.noaa.gov/data/oceans/archive/arc0216/0253808/

10 References

1. WMO (2017). *WMO Guidelines on the Calculation of Climate Normals* (WMO-No. 1203). Geneva: World Meteorological Organization.
2. Copernicus Climate Change Service (C3S). *Climate Indicators: About the data*. climate.copernicus.eu/climate-indicators/about-data